Integration of Open OnDemand with the Jobstats Job Monitoring Platform

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GOOD Conference at Harvard University
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Why the Jobstats Job Monitoring Platform?

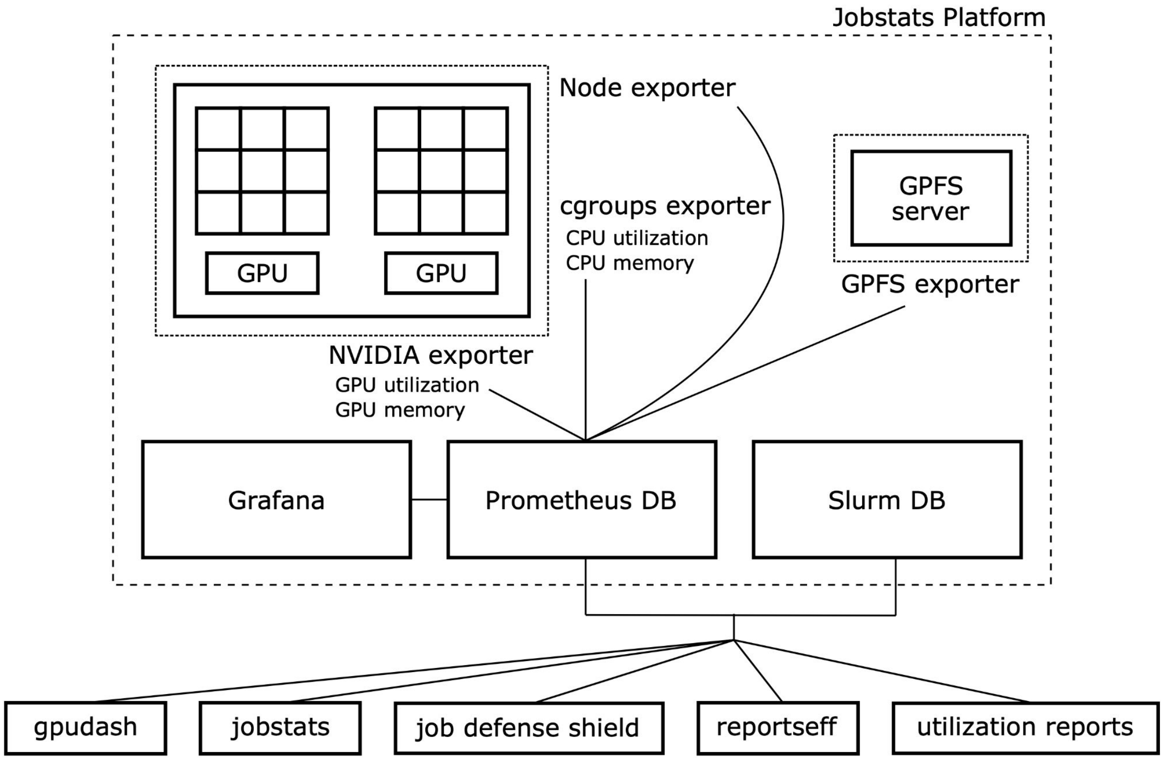
Multinode CPU fragmentation (96 nodes, 1 CPU-core per node)

```
  JobID        State                Start      Elapsed NNodes NCPUS  JobName
------- ----------- ------------------- ------------ ------ ----- --------
1929990      TIMEOUT 2025-03-10T14:19:24     02:05:21     96    96 coffee-+
1930000      TIMEOUT 2025-03-10T14:30:29     05:35:16      5   448 merger-+
1930364 CANCELLED + 2025-03-10T18:28:21     01:08:49     16  1536 magneta+
```

GPU job without GPU-enabled software (CPU-only PyTorch)

```
pysocks             pkgs/main/linux-64::pysocks-1.7.1-py312h06a4308_0
python              pkgs/main/linux-64::python-3.12.9-h5148396_0
pytorch             pytorch/linux-64::pytorch-2.5.1-py3.12_cpu_0
pytorch-mutex       pytorch/noarch::pytorch-mutex-1.0-cpu
pyyaml              pkgs/main/linux-64::pyyaml-6.0.2-py312h5eee18b_0
```

Design of the Jobstats Job Monitoring Platform



```
$ jobstats 46926519
================================================================================
                              Slurm Job Statistics
================================================================================
         Job ID: 46926519
   User/Account: u12345/ee
       Job Name: interactive
          State: COMPLETED
          Nodes: 1
      CPU Cores: 4
     CPU Memory: 4GB (1GB per CPU-core)
           GPUs: 4
  QOS/Partition: gpu-short/gpu
        Cluster: della
     Start Time: Thu Apr 13, 2023 at 10:59 AM
       Run Time: 02:05:28
     Time Limit: 02:00:00

                              Overall Utilization
================================================================================
  CPU utilization  [||||                                            8%]
  CPU memory usage [||||||||||||||||||||||||||||||||||||||||||||||| 92%]
  GPU utilization  [                                                0%]
  GPU memory usage [|                                               2%]
```

```
                         Detailed Utilization
================================================================================
  CPU utilization per node (CPU time used/run time)
      della-l03g11: 00:39:04/08:21:52 (efficiency=7.8%)
  CPU memory usage per node - used/allocated
      della-l03g11: 3.7GB/4.0GB (941.9MB/1.0GB per core of 4)
  GPU utilization per node
      della-l03g11 (GPU 0): 0% <--- GPU was not used
      della-l03g11 (GPU 1): 0% <--- GPU was not used
      della-l03g11 (GPU 2): 0% <--- GPU was not used
      della-l03g11 (GPU 3): 0% <--- GPU was not used
  GPU memory usage per node - maximum used/total
      della-l03g11 (GPU 0): 3.2GB/80.0GB (4.0%)
      della-l03g11 (GPU 1): 1.9GB/80.0GB (2.4%)
      della-l03g11 (GPU 2): 694.4MB/80.0GB (0.8%)
      della-l03g11 (GPU 3): 694.4MB/80.0GB (0.8%)

                                Notes
================================================================================
  * This job did not use 4 of the 4 allocated GPUs. Please resolve this before
    running additional jobs. Wasting resources causes your subsequent jobs to
    have a lower priority. Is the code capable of using multiple GPUs? Please
    consult the documentation for the software. For more info:
      https://researchcomputing.princeton.edu/KB/gpu-computing
```

```
$ jobstats 1749336
================================================================================
                              Slurm Job Statistics
================================================================================
         Job ID: 1749336
   User/Account: u12345/astro
       Job Name: mlogit_bs
          State: CANCELLED
          Nodes: 1
      CPU Cores: 12
     CPU Memory: 48GB (4GB per CPU-core)
  QOS/Partition: long/all
        Cluster: adroit
     Start Time: Tue Apr 18, 2023 at 10:29 PM
       Run Time: 1-10:46:22
     Time Limit: 2-00:00:00
                              Overall Utilization
================================================================================
  CPU utilization  [||||                                            8%]
  CPU memory usage [||                                              4%]
                              Detailed Utilization
================================================================================
  CPU utilization per node (CPU time used/run time)
      adroit-h11n4: 1-10:40:39/17-09:16:24 (efficiency=8.3%)
  CPU memory usage per node - used/allocated
      adroit-h11n4: 2.0GB/48.0GB (173.4MB/4.0GB per core of 12)
```

```
                                     Notes
================================================================================
  * The overall CPU utilization of this job is 8%. This value is low compared
    to the target range of 90% and above. Please investigate the reason(s) for
    the low efficiency. For instance, have you conducted a scaling analysis?
    For more info:
      https://researchcomputing.princeton.edu/KB/cpu-utilization

  * The CPU utilization of this job (8%) is equal to 1 divided by the number
    of allocated CPU-cores (1/12=8%). This suggests that you may be running a
    code that can only use 1 CPU-core. If this is true then allocating more
    than 1 CPU-core is wasteful. Please consult the documentation for the
    software to see if it is parallelized. For more info:
      https://researchcomputing.princeton.edu/KB/parallel-code

  * This job only used 4% of the 48GB of total allocated CPU memory. For
    future jobs, please allocate less memory by using a Slurm directive such
    as --mem-per-cpu=1G or --mem=3G. This will reduce your queue times and
    make the resources available for your other jobs. For more info:
      https://researchcomputing.princeton.edu/KB/memory

  * For additional job metrics including metrics plotted against time:
      https://myadroit.princeton.edu/pun/sys/jobstats
```

Jobstats Features

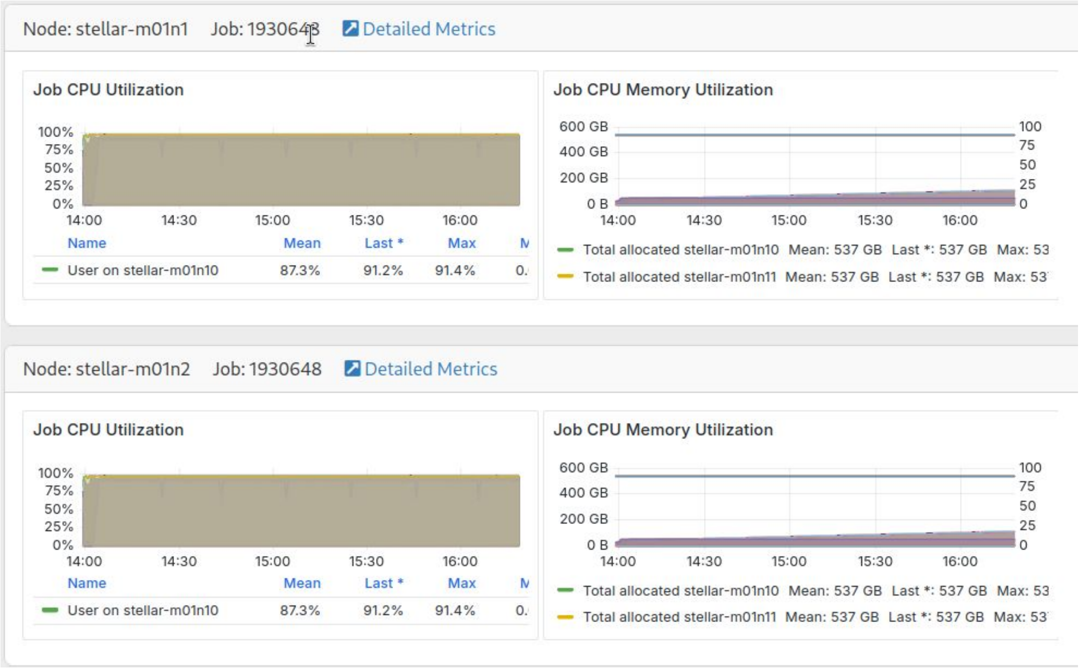
- **Automatically cancel GPU jobs at 0% utilization**
- Identify and/or email users with the most GPU-hours at 0% utilization
- Identify and/or email users that are over-allocating CPU memory
- and 10 other instances of underutilization

```
Hi Alan (u12345),

The jobs below have been cancelled because they ran for 2 hours at 0% GPU
utilization:

    JobID    Cluster  Partition    State    GPUs-Allocated GPU-Util  Hours
   60131148   della      llm     CANCELLED         4           0%        2
   60131741   della      llm     CANCELLED         4           0%        2

Replying to this automated email will open a support ticket with Research
Computing.
```

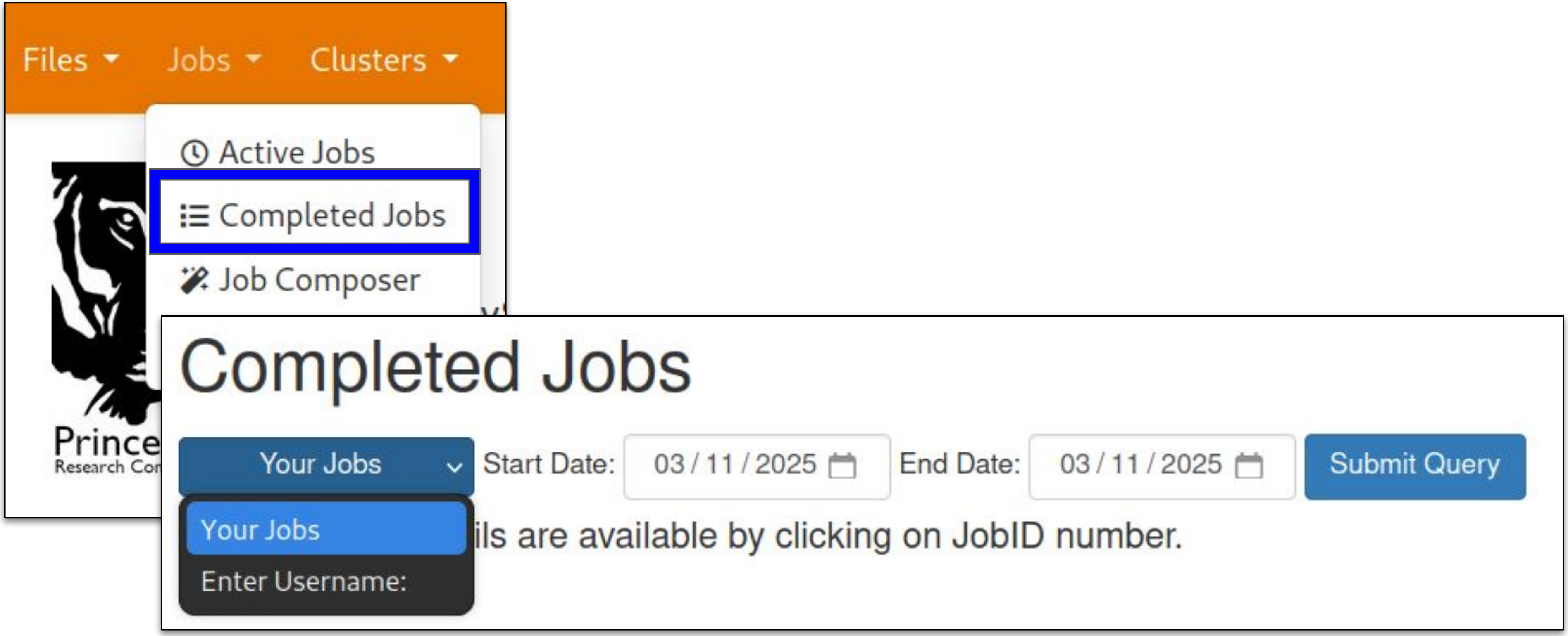
github.com/PrincetonUniversity/jobstats

Open OnDemand Basic Grafana Integration

```yaml
---
v2:
  metadata:
    title: Super Cluster
  ....
  custom:
    grafana:
      host: https://ood.server/grafana
      orgId: 1
      dashboard:
        name: ondemand-clusters
        uid: ZALk3fG9a
        panels:
          cpu: 7
          memory: 9
      labels:
        cluster: cluster
        host: node
        jobid: JobID
```



Open OnDemand Completed Jobs



Open OnDemand Completed Jobs

Job 1928348 ("2xIC") [click for full stats]

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Memory and CPU Efficiency

Node	Memory Used/Allocated	Memory Per Core	CPU Time Used/Allocated	CPU Efficiency	Cores
stellar-m01g2	4.5 GB / 14.6 GB (30.5%)	2.2 GB / 7.3 GB	3-23:48:19 / 4-00:10:06	99.6%	2
Total:	4.5 GB / 14.6 GB (30.5%)		3-23:48:19 / 4-00:10:06	99.6%	2

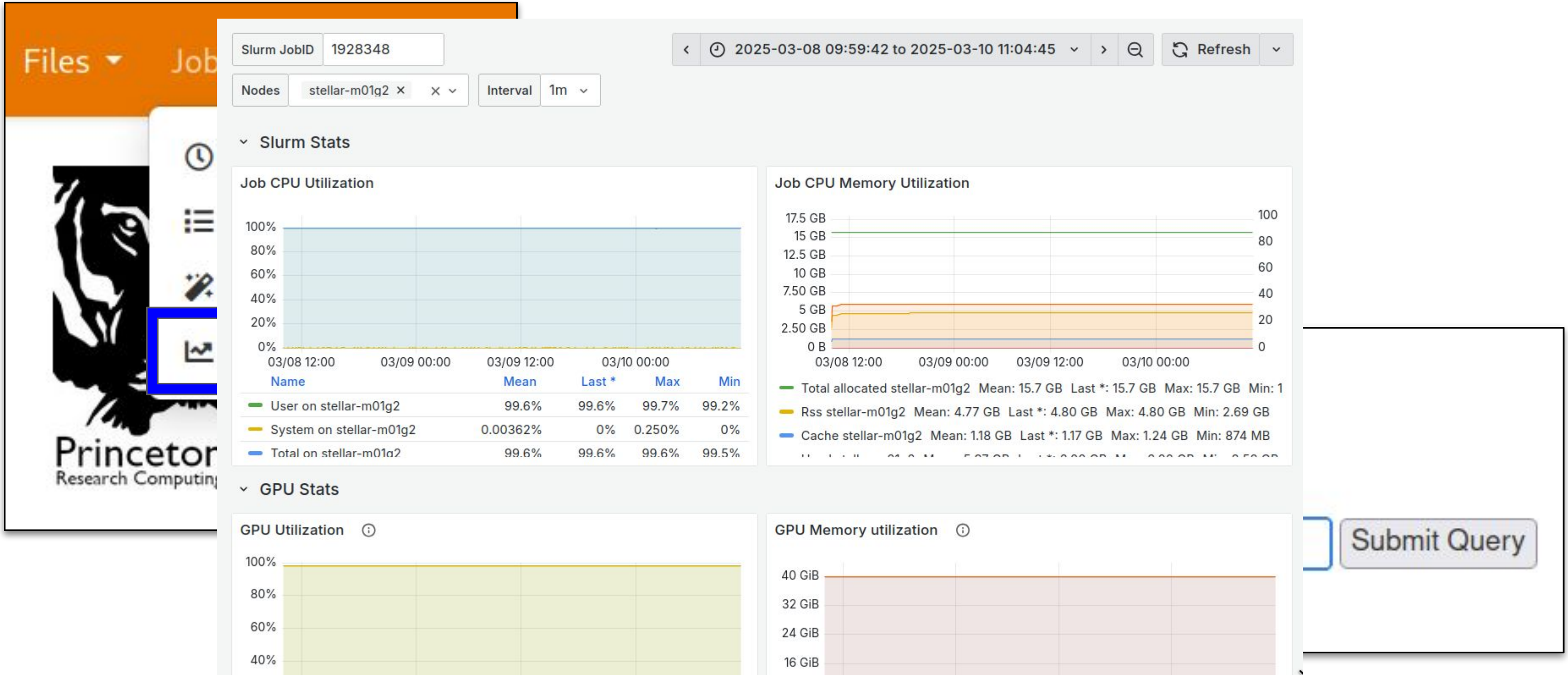
GPU Efficiency

Node	GPU	GPU Max Memory Used/Total	Average GPU Utilization
stellar-m01g2	#0	11.4 GB / 40 GB (28.5%)	98.0%
stellar-m01g2	#1	11 GB / 40 GB (27.5%)	98.0%
Total:	2 GPU(s)	22.4 GB / 80 GB (28%)	98.0%

Close

OPEN

Open OnDemand Helper App



Summary

- `jobstats` command shows CPU/GPU utilization and CPU/GPU memory usage
- **Automatically cancel GPU jobs at 0% utilization**
- Identify and/or send email alerts to users for underutilization
- Used by tens of institutions throughout the world
- Jobstats integrates with OOD making it easy to use

`github.com/PrincetonUniversity/jobstats`